

AB POWER ENGINEERING

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ENGINE COOLING AND LUBRICANT SYSTEM

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1. System used to maintain the optimum temperature of the engine low enough so it does not reach the melting point of metals.

- a. Engine cooling system
- b. Engine air-intake system
- c. Engine air-exhaust system
- d. None of the above

2. Which of the following is a result when the cooling system of an engine fails?

- a. Oil at the engine will oxidize.
- b. Distortion of engine metal parts.
- c. Piston will difficult to function.
- d. All of the above
- e. None of the above

3. During cooling, the percentage amount of heat needed to be removed from the engine is ____.

- a. 10%
- b. 30%
- c. 70%
- d. All of the above

4. Which is not a type of engine cooling system?

- a. Air cooling
- b. Water cooling
- c. Oil cooling
- d. None of the above

5. Too much removal of heat during cooling will ____ the thermal efficiency of the engine.

- a. increase
- b. reduce
- c. not change
- d. None of the above

6. Cooling of engine is slow during the starting in order for the engine to reach its operating ____ in a short time.

- a. pressure
- b. temperature
- c. speed
- d. None of the above

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7. This reduces the temperature of the engine by means of a stream of air only from the surroundings.

- a. Air cooling
- b. Water cooling
- c. Water and air cooling
- d. None of the above

8. Heat transfer fins are provided at the cylinder block and cover of the engine to ____.

- a. strengthen the machine part to be cooled
- b. increase the contact area for cooling
- c. increase the weight of the engine
- d. All of the above

9. This is provided for stationary air-cooled engines to effect on cooling.

- a. Pump
- b. Fan
- c. Fins
- d. All of the above

10. Which of the following is the factor that determine the amount of heat during air cooling of engine?

- a. Area of fin surfaces
- b. Velocity and amount of cooling air
- c. Temperature of fins and cooling air
- d. All of the above

11. Which of the following are the advantages of air-cooled engines?

- a. Light in weight
- b. Less care and maintenance
- c. Cheaper to manufacture
- d. No damage from frost cracking of cylinder jackets or radiator tubes
- e. All of the above
- f. Two of the above

12. Type of water-cooling system for engine.

- a. Direct or non-return system
- b. Thermo-siphon system
- c. Hopper system
- d. Pump/Force circulation system
- e. All of the above

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13. Air cooling are usually adopted for ____.

- a. single-cylinder engine
- b. four-cylinder engine
- c. six-cylinder engine
- d. None of the above

14. Engines usually equipped with radiator are ____.

- a. air cooled
- b. water cooled
- c. oil cooled
- d. None of the above

15. Purpose of lubrication engine.

- a. To reduce friction.
- b. To help in cooling engine.
- c. To provide sealing effect between the piston and cylinder liner.
- d. To clean the engine.
- e. All of the above
- f. Three of the above

16. Used to cool down the hot water received from the engine by using a fan that blows the air through its fins.

- a. Thermostat
- b. Radiator
- c. Pump
- d. All of the above

17. Which of the following is not a part of a forced-circulation water-cooling system?

- a. Water pump
- b. Fan
- c. Radiator
- d. Pressure cup
- e. Fan belt
- f. Water jacket
- g. Thermostat valve
- h. Temperature gauge
- i. Hose pipe
- j. All of the above
- k. None of the above

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18. Which of the following are qualities of good lubricants?

- a. Sufficient viscosity
- b. Stable under changing temperatures
- c. Keep lubricated parts clean
- d. Not corrosive
- e. All of the above
- f. Two of the above

19. Engine part that requires lubrication.

- a. Cylinder wall
- b. Piston and piston pins
- c. Crankshaft
- d. Connecting rod bearings
- e. Camshaft bearings
- f. Valves and valve operating mechanism
- g. Cooling fans
- h. Water pump
- i. All of the above
- j. Four of the above

20. Used to remove dirty elements in the oil in an effective way which is made of cloth, paper, felts, and wire screen.

- a. Air filter
- b. Gas filter
- c. Oil filter
- d. None of the above

21. Used to prevent building up of pressure in the crankcase of the engine.

- a. Relief valve
- b. Crankcase breather
- c. Oil pressure gauge
- d. None of the above

22. Common lubrication systems in multiple-cylinder engines like in tractors.

- a. Splash system
- b. Forced-feed system
- c. Combination of splash and forced-feed system
- d. All of the above
- e. Two of the above

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23. A fluid that circulates through a cooling system of water-cooled engine to absorb excess heat preventing it from overheating and maintaining an optimal operating temperature.

- a. Oil
- b. Lubricant
- ☒ c. Coolant
- d. None of the above

24. Purpose of providing coolant to engine.

- a. Prevent freezing of cooling fluid during cold temperature
- b. Protects internal engine parts from corrosion
- c. Protect internal parts of the engine from rust
- ☒ d. All of the above

25. Functions of coolant.

- a. Absorbs heat
- b. Lowers the freezing point of fluid
- c. Increases the boiling point of fluid
- d. Lubricates and protects corrosion
- ☒ e. All of the above

26. A coolant is a ____.

- a. colored water
- b. glycol-based fluid
- ☒ c. diluted hydroxide
- d. None of the above

27. What are the key components of engine cooling system?

- a. Coolant
- b. Water pump
- c. Radiator
- d. Thermostat
- ☒ e. Cooling fan
- f. All of the above

28. Which of the following regulates the flow of coolant ensuring the flow of proper temperature?

- a. Radiator
- b. Water pump
- ☒ c. Thermostat
- d. Cooling fan
- e. None of the above

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29. Types of coolant

- a. Inorganic additive technology
- b. Organic Acid technology
- c. Hybrid Acid technology
- ☒ d. All of the above

30. Type of coolant that provide quick but sacrificial form of corrosion protection with typically green color.

- ☒ a. Inorganic additive technology
- b. Organic Acid technology
- c. Hybrid Acid technology
- d. All of the above

31. Type of coolant that offers longer-lasting or "extended life protection with often color of red, orange, yellow or purple.

- a. Inorganic additive technology
- ☒ b. Organic Acid technology
- c. Hybrid Acid technology
- d. All of the above

32. It is recommended not to mix different types of coolants, as it can dilute the additives and _____ their effectiveness and protective qualities.

- ☒ a. instantly destroy
- ☒ b. significantly reduce
- c. significantly improve
- d. None of the above

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